

Dr. Arnab Kanti Karan

PhD in Physics | Materials Science &
Optoelectronic Devices | Perovskite & Organic
Materials | Thin Film Technology

CONTACT

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EDUCATION

PhD in Physics
Jadavpur University
2019 – 2025

MSc in Physics
Techno India University
2015 – 2017 | CGPA 8.08

CORE SKILLS

- Optical & optoelectronic characterization
- Electrical & interfacial characterization
- Thin-film & nanomaterial synthesis
- XRD, SEM/EDX, AFM, Raman, XPS, UPS (With instrument handling and analysis)
- DFT modelling (Quantum Espresso)
- Python-based chemical structure Optimisation

PROFILE

Doctoral researcher in Condensed Matter Physics with a research focus on the optical, electrical and structural characterisation of thin-film photovoltaic and optoelectronic devices, including dye-sensitised solar cells and lead-free perovskite-based devices. Experienced in correlating photon management, charge transport and interfacial/barrier phenomena with device performance (spectral response, fill factor, power conversion efficiency, absorption coefficient), and in synthesising and structurally/spectroscopically characterising nanomaterial- and nanoparticle-modified thin films. Author of 19 peer-reviewed publications as corresponding author and 11 conference contributions, with hands-on experience across a broad range of structural, optical, electrical and spectroscopic characterisation instrumentation.

RELEVANT EXPERIENCE

Doctoral Researcher — Condensed Matter Physics Research Centre, Jadavpur University
2019 – 2025

While trained as an experimental physicist, my day-to-day research depended heavily on simulation and experimental work:

- Wrote and ran Python-based routines within Quantum Espresso to perform DFT electronic-structure calculations to support experimental findings.
- Developed Fortran-based code for Arduino for instrument control and an automated data acquisition device for characteristics experiments.
- Documented methods and results clearly for 19 peer-reviewed publications – dedication to completion.
- Guided 8 M.Sc. students through their dissertation projects, and continue to mentor 5 Ph.D. research scholars in my lab as a senior researcher - experience in team collaboration, task delegation and peer support.

Guest Teacher — Department of Physics, Ramnagar College ([Link](#))
2023 – 2026

- Delivered undergraduate lectures in physics, covering core topics such as classical mechanics, electromagnetism, condensed matter/solid-state physics, Python, Fortran, analogue-digital circuits, etc.
- Supervised undergraduate physics laboratory sessions, guiding students through experimental setup, measurement techniques, and data analysis.
- Prepared lecture materials, assignments, and assessments aligned with the department's curriculum.
- Mentored undergraduate students on coursework and academic queries, drawing on research experience in materials characterisation and device physics.
- Contributed to departmental academic activities alongside ongoing doctoral research responsibilities.

RESEARCH EXPERIENCE

Doctoral Research — Materials–Device Interfaces & Photovoltaic Characterisation
Condensed Matter Physics Research Centre, Jadavpur University | 2019 – 2025

- **Thesis:** Influence of Nanoparticles and Carbon Nanotubes on Electrical Characteristics of Natural and Organic Dye-based Cells — studied how nanoparticle and carbon-nanotube incorporation modifies charge transport, light absorption and stability in organic dye-based photovoltaic devices.
- **Extra research activities:** lead-free perovskite-based optoelectronic devices; rGO–polymer nanocomposites for gas sensing and high-dielectric applications; perovskite-based devices for memory-switching applications.

- SCAPS device simulation
- C, Fortran, Arduino, DBMS
- Arduino-based instrumentation

DEVELOPED WEBSITE

isdrcju.in

temp.control

AT A GLANCE

19 peer-reviewed publications

11 conference presentations

- **Mentoring:** guided 8 M.Sc. students through their dissertation projects, and continue to mentor 5 Ph.D. research scholars in the lab as a senior researcher.

KEY PROJECTS

DFT-Based Electronic Structure Modelling (Fortran / Quantum Espresso)

Used Python-based DFT routines to model electronic structure in perovskite and dye-based semiconductor materials, correlating computational predictions with experimental measurements.

Conference Website Development

Collaborated with my juniors as part of a team to design and develop the official website for [\[isdrcju.in\]](http://isdrcju.in), while coordinating with fellow team members to ensure accurate scheduling, speaker information, and timely updates ahead of and during the event; also helped troubleshoot website issues to ensure a smooth experience for attendees.

Arduino-Based Competitive Project Mentoring

- Single-handedly mentored undergraduate students on Arduino programming for competitive science projects, guiding them through the design and development of applications such as a digital millisecond watch and automatic street-light controller, while supporting their understanding of circuit logic, sensor integration, and code debugging to help them complete and present their projects; they obtained 1st prize.
- Mentored a postgraduate student in developing a website ([Link](#)) to automatically control temperature and humidity for household applications using an Arduino-ESP32 programming module. This device was successfully implemented in our lab for experimental purposes.
- We developed an Arduino-based timer-controlled spin-coater machine with controllable R.P.M. in our lab.

FULL LIST OF PUBLICATIONS

- A.K. Karan*, D. Sahoo, N.B. Manik. Enhanced photoelectric performance of sunset yellow dye-zinc oxide composed solar cells. *Materials Letters*, 402, 2025.
- D. Sahoo*, A.K. Karan, N.B. Manik. MWCNT-Enhanced Cs₂SnCl₆ Perovskite for Improved Charge Transport in a Smartphone-Interfaced UV Photodetector. *ACS Applied Electronic Materials*, 7(9), 4391–4402, 2025.
- S. Rakshit*, A.K. Karan, N.B. Manik. Investigation of the effect of SWCNT on semiconducting properties of turmeric dye-based Schottky device. *J. Materials Science: Materials in Electronics*, 36, 821, 2025.
- A. Samanta, M. Shit, A.K. Karan, N.B. Manik, C. Sinha*, S. Khanra. Zn(II)-based 2D coordination polymer bridged by isophthalate and dipyritylsulphide. *J. Molecular Structure*, 1321(2), 2024.
- S. Rakshit*, A.K. Karan, N.B. Manik. Enhanced Electrical Transport Properties of Beetroot Dye-Based Thin Film in Presence of TiO₂ Nanoparticles. *J. Electronic Materials*, 53, 3914–3925, 2024.
- A.K. Karan*, D. Sahoo, S. Sen, S. Rakshit, S. Bhunia, N.B. Manik. Estimation of activation energy of Tartrazine dye based natural organic device. *AIP Conference Proceedings*, 3067, 020023, 2024.
- A.K. Karan*, D. Sahoo, S. Sen, N.B. Manik. Modification of barrier height inhomogeneity in the presence of TiO₂ nanoparticles on Carmoisine dye-based Schottky device. *Surfaces and Interfaces*, 46, 2024.
- M. Shit, A.K. Karan, S. Maity, A.M.Z. Slawin, N.B. Manik, B. Dutta, C. Sinha*. Succinato-bridged Cd(II)-nicotinyldihydrazone 3D coordination polymer. *J. Coordination Chemistry*, 77, 142–154, 2024.

- M. Shit, S. Halder, K. Manna, A.K. Karan, A. Samanta, N.B. Manik, S. Pal, K. Jana, C. Sinha*. Mn(II) 3D coordination framework with mixed 5-aminoisophthalato and pyridyl-isonicotinoyl hydrazone bridges. *ACS Applied Polymer Materials*, 6, 2637–2648, 2024.
- A.K. Karan*, D. Sahoo, N.B. Manik. Investigating the effects of TiO₂ nanoparticles on the barrier inhomogeneity of brilliant-blue fruit dye-based solar cell. *Current Applied Physics*, 59, 95–104, 2023.
- A.K. Karan*, D. Sahoo, N.B. Manik. Enhanced electrical conductivity and charge conduction mechanisms in nano-cubical Sunset Yellow dye incorporated with TiO₂ nanoparticles. *Physica B: Condensed Matter*, 674, 2023.
- D. Sahoo*, P. Sengupta, A.K. Karan, N.B. Manik. Improvement in conductivity of lead-free CH₃NH₃SnI₃ perovskite thin film using multi-walled carbon nanotubes as a transporter. *Surfaces and Interfaces*, 10, 1060, 2023.
- A.K. Karan*, D. Sahoo, S. Sen, N.B. Manik. Electrical conduction mechanism of carmoisine dye-based natural organic device. *Indian Journal of Physics*, 98, 577–583, 2024.
- A.K. Karan*. Effect of TiO₂ nanoparticle on Richardson Constant and Barrier Height of Tartrazine Dye based Schottky Device. *Discover Materials*, 3, 2023.
- D. Sahoo*, A.K. Karan, N.B. Manik. Influence of SWCNT on the Electrical Behaviour of an Environmentally Friendly CH₃NH₃SnI₃ Perovskite-Based Optoelectronic Schottky Device. *ACS Applied Electronic Materials*, 2023.
- D. Sahoo*, A.K. Karan, Z. Mallick, N.B. Manik. Synthesis and complex impedance analysis of nano-cubic CH₃NH₃SnI₃ perovskite for the development of an optoelectronic lead-free Schottky diode. *Materials Science in Semiconductor Processing*, 155, 2023.
- D. Sahoo*, A.K. Karan, N.B. Manik. Electrical charge transport properties of caesium tin chloride perovskite microrods. *Materials Letters*, 339, 2023.
- M. Shit, A.K. Karan, D. Sahoo, N.B. Manik, B. Dutta, C. Sinha*. Strategy for the improvement of electrical conductivity of a 3D Zn(II)-coordination polymer doubly bridged by mesaconato and pyridyl-isonicotinoyl hydrazide based Schottky diode device. *New Journal of Chemistry*, 47, 5922–5929, 2023.
- A.K. Karan*, S. Bhunia, N.B. Manik. Study on the Conductivity of a Sunset Yellow Dye-Based Natural Organic Device. *J. Electronic Materials*, 51, 7156–7163, 2022.

CONFERENCE & SEMINAR PRESENTATIONS

- Highly Efficient Photo-Electric Behaviour of Carnosine-based Synthetic DSSC incorporated with ZnO Nanoparticles — 32nd CMDAYS, 2024
- Investigating the Influence of TiO₂ Nanoparticle Doping on Polaron Hopping Distance in Tartrazine Dye Cells — 4th ICMS, 2024
- Improved Conductivity through Barrier Inhomogeneity Modification via TiO₂ Nanoparticle Integration in Sunset Yellow Dye-Based Schottky Diode — 31st CMDAYS, 2023
- Modification of Richardson Constant and Interfacial Potential in the Presence of TiO₂ Nanoparticles on Carmoisine Dye-Based Device — 7th ICONN, 2023
- Estimation of Richardson Constant and Barrier Inhomogeneity in the Presence of TiO₂ Nanoparticles on Tartrazine Dye-Based Organic Schottky Device — 30th CMDAYS, 2022
- Evaluation of Richardson Constant of Fruit Dyes Using Carmoisine and Tartrazine — IEMPHYS, 2022
- Effect of TiO₂ Nanoparticle on Richardson Constant of Sunset Yellow Dye-Based Cell — ICNT, 2022
- Estimation of Activation Energy of Tartrazine Dye-Based Natural Organic Device — 4th CTMSE, 2022
- Study on the Conductivity of Natural Organic Dye-based Device Using Tartrazine Dye — ICRTGC, 2021
- Estimation of Richardson Constant for Natural Organic Dye-Based Cells using Orange-Lemon and Apple-Green — 3rd CTMSE, 2021
- Effect of TiO₂ on Malachite Green Dye-based Organic Diode — National Seminar on New Directions in Physical Sciences, 2020

ADDITIONAL

Extracurricular: Senior Diploma in Classical Music; Junior Diplomas in Rabindra Sangeet and Nazrul Geeti; elocution/recitation; B.Sc. General (Computer Science), Kharagpur College.

Interests: gardening, photography, wildlife camping, nature exploration.